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Data Integration as a balance point between resilience, meaning, and democracy

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Who I am

A Boomer, born 1961, married with two children

Work and more

- 1987 – 1998, Tecsiel: Researcher in Natural Language Processing, Knowledge Representation, Machine Learning and Software Development Methodologies
- 1998 – 2018, Oracle: Account Manager, Innovation and Business Advisor
- From 2019, Denodo: Data Evangelist
- Passionate philosopher, Food Blogger (Tra Pignatte e Sgommarelli), passionate about nonverbal communication (De Corporis Voce) and scholar and activist of Villa Ada in Roma

Education

- 1987 - Bachelor's degree with honors in demographic statistics, methodological focus, from the University of Rome "La Sapienza"
- 1989 - Summer School on Machine Learning, University of Turin

Denodo

A Leader in Data Management for 25 years

Denodo, leader in Data Management

Since 1999, focused on a logical approach to data management

1000+ customers

including many F500 and G2000 companies, across all market sectors.

TOYOTA · BNP PARIBAS · sanofi · AirEuropa · BHP · TotalEnergies · ABN·AMRO · insiel · Istat

Global presence

25 offices in 20 countries

800+ employees

250+ Partners

Active in the world

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Recognised as Leader by Analysts

GARTNER
Leader in the 2024 Gartner® Magic Quadrant™ for Data Integration Tools

Fifth year in a row

FORRESTER
Leader in The Forrester Wave™: Enterprise Data Fabric, Q2 2022

Second year in a row

and the clients agree

GARTNER
Customers' Choice in 2024 Gartner Peer Insights™ "Voice of the Customer": Data Integration Tools Report

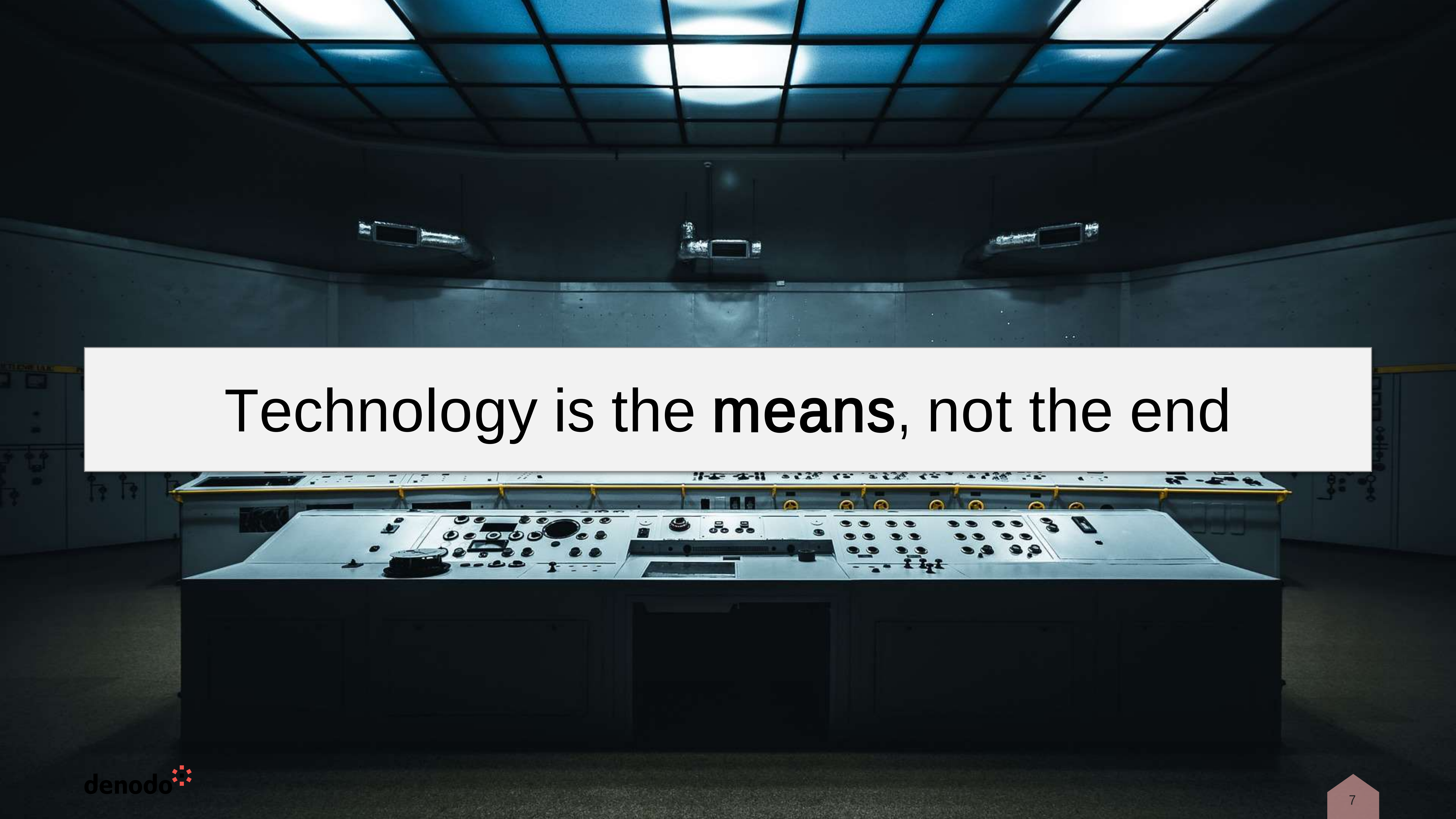
Fourth year in a row

Some premises, before starting

We must first understand the role of technology and then we can use it.

“Technology changes, but the laws of economics don’t!”

Carl Shapiro e Hal R. Varian - *“Information Rules: A Strategic Guide to the Network Economy”* - 1998


Technology is the **means**, not the end

The goal, for any company or organization, is always a **value**, which in most cases is the **competitive advantage** derived from it

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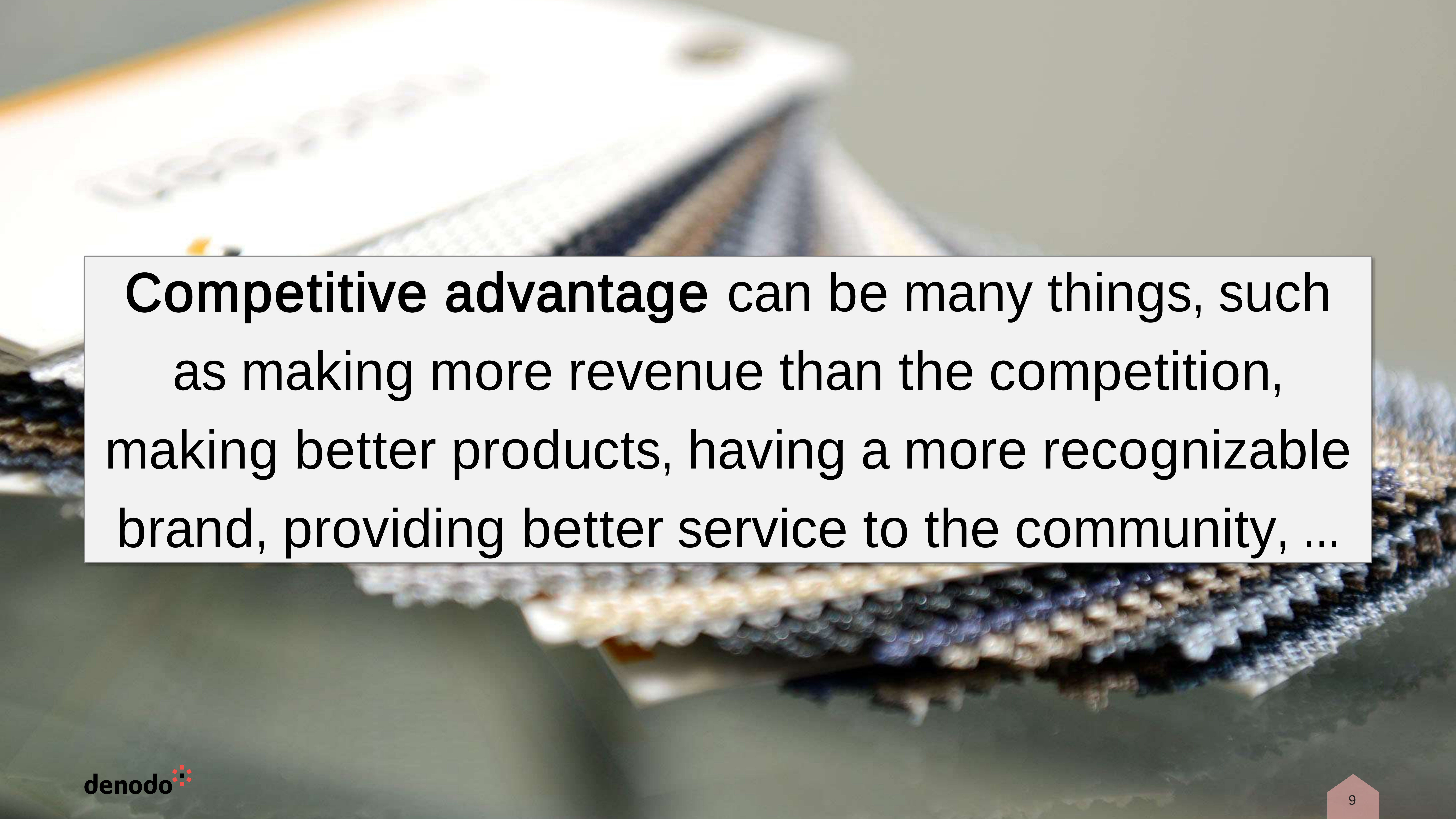


Competitive advantage can be many things, such as making more revenue than the competition, making better products, having a more recognizable brand, providing better service to the community, ...

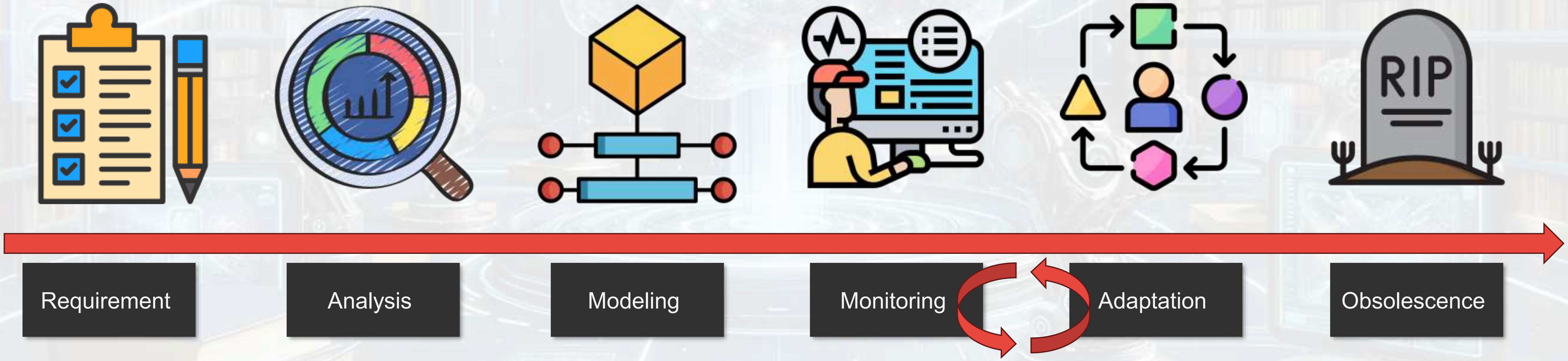
Experiments, projects, proofs of concept, should never start to test a technology, but only and when there is a related **business objective** that justifies it and can be **measured**

Representing a concept, in time and space

Any concept is semantically placed in a specific context, characterized by temporal, social and cultural elements, which inevitably shape its meaning, adapting it to the changes that this context undergoes

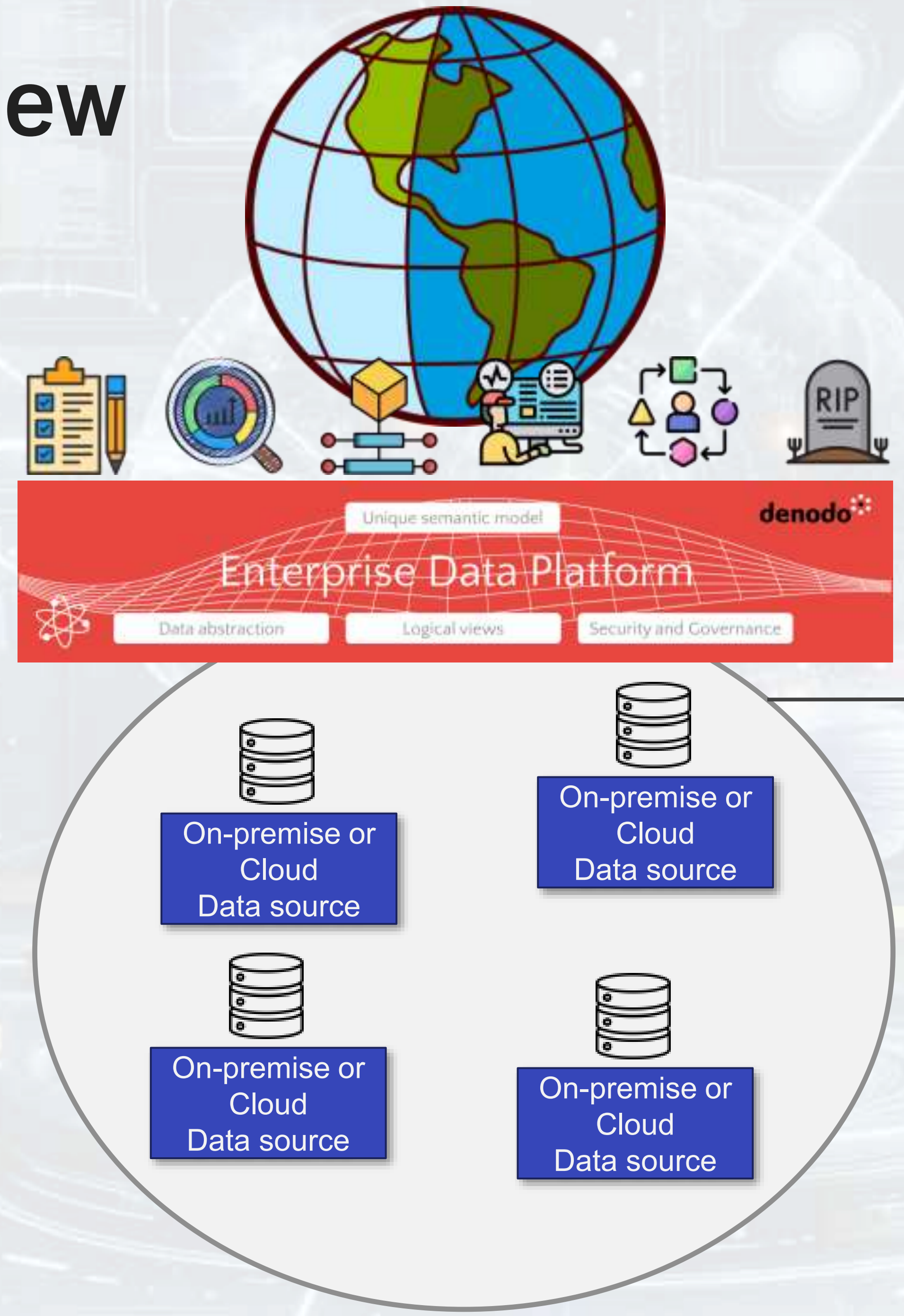
Representing a concept, in time and space

Every **concept**, especially if it is articulated and complex, has some intrinsic characteristics that define what it is, characteristics that typically change during its existence, from the birth of the concept itself to its eventual obsolescence



In the digital world every **representation** is nourished by data, it is therefore necessary to define the different levels that allow this representation to be gradually extended, so that it is increasingly richer and more complete, leading us to **Augmented Concept View**

Augmented Concept View

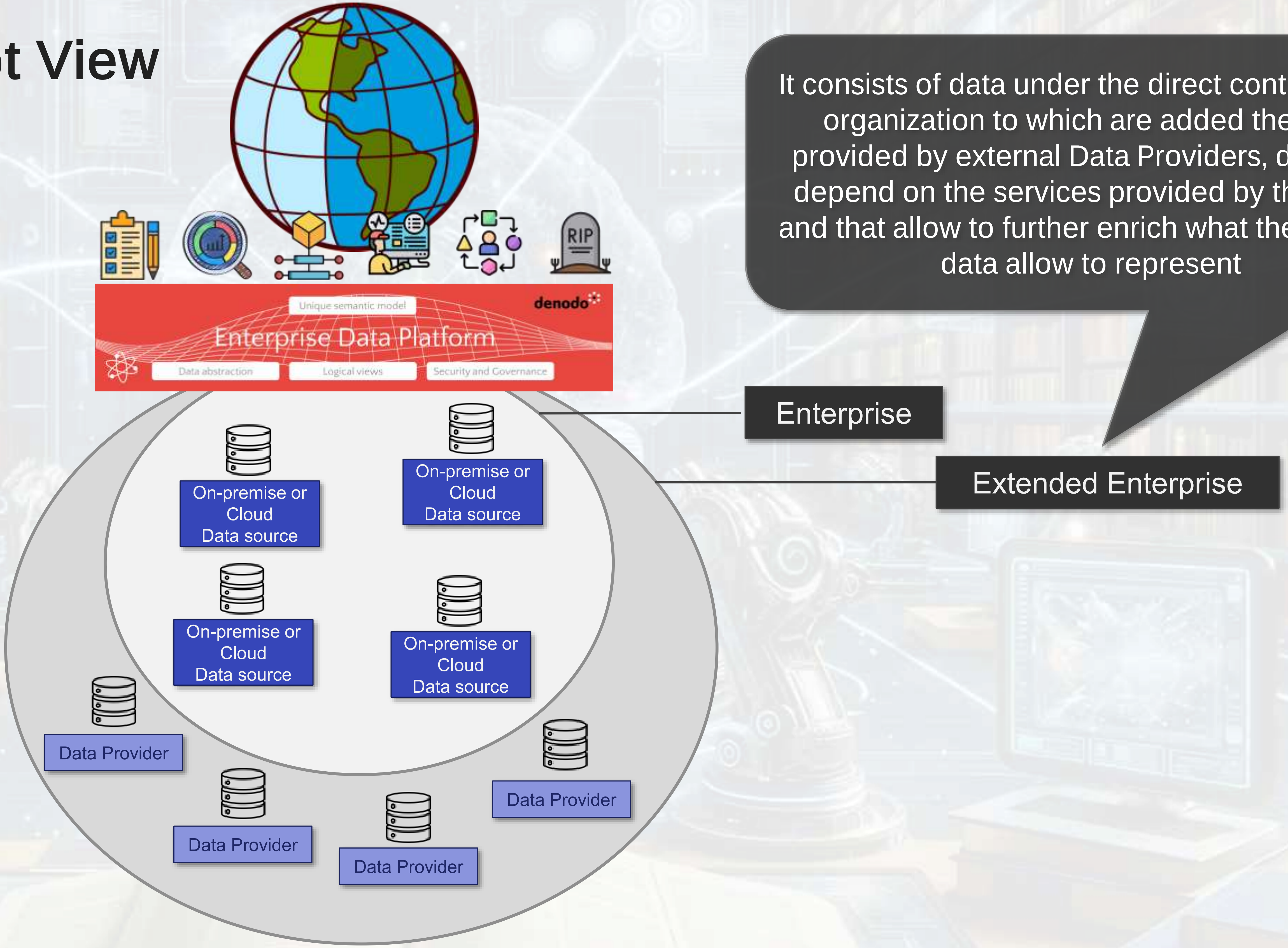


It consists of data under the direct control of the organization, data for which the organization has ownership and responsibility, regardless of whether they are on-premises or in the cloud

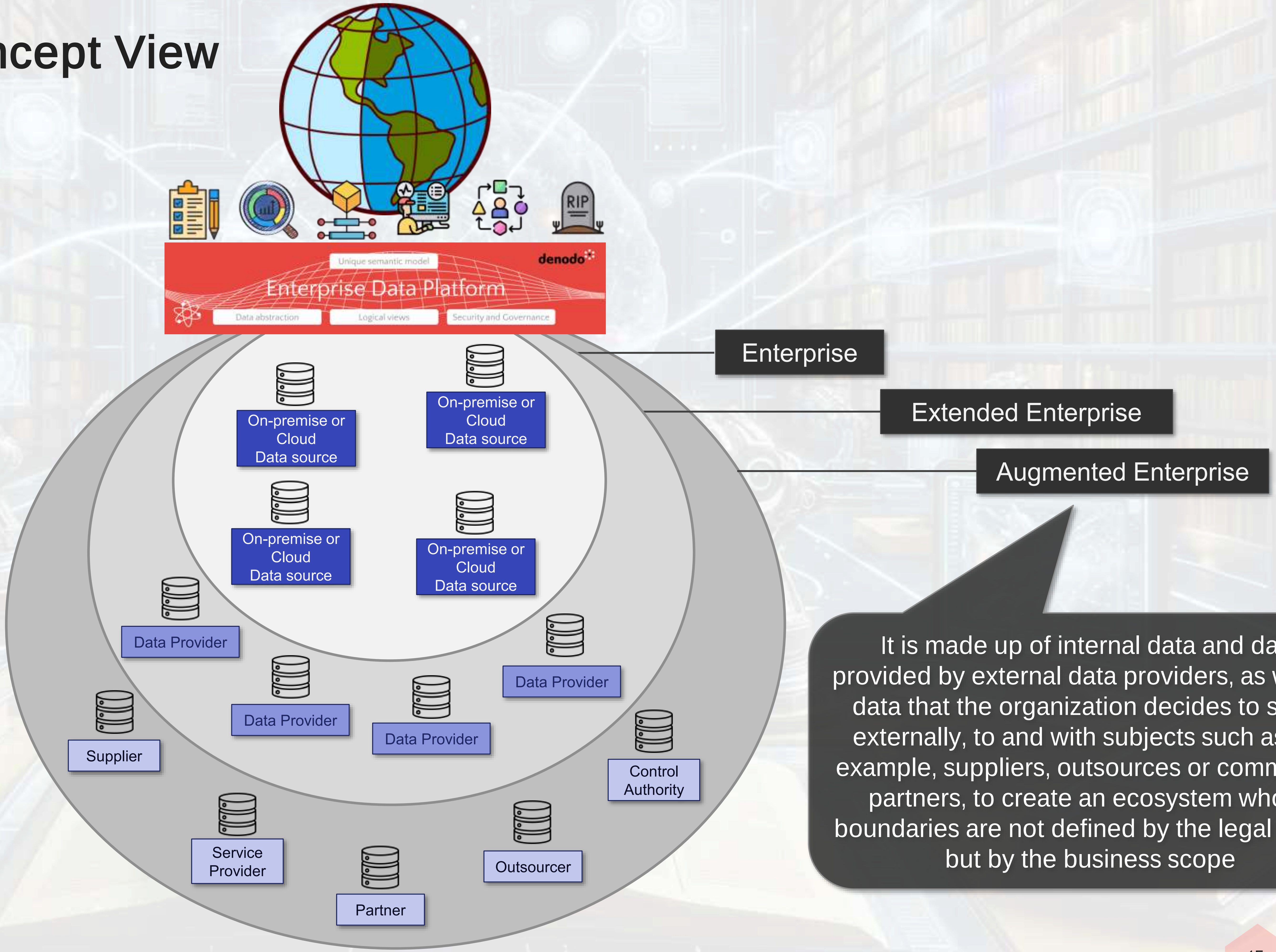
Enterprise

Augmented Concept View

It consists of data under the direct control of the organization to which are added the data provided by external Data Providers, data that depend on the services provided by the latter and that allow to further enrich what the internal data allow to represent



Augmented Concept View



It is made up of internal data and data provided by external data providers, as well as data that the organization decides to share externally, to and with subjects such as, for example, suppliers, outsources or commercial partners, to create an ecosystem whose boundaries are not defined by the legal scope but by the business scope

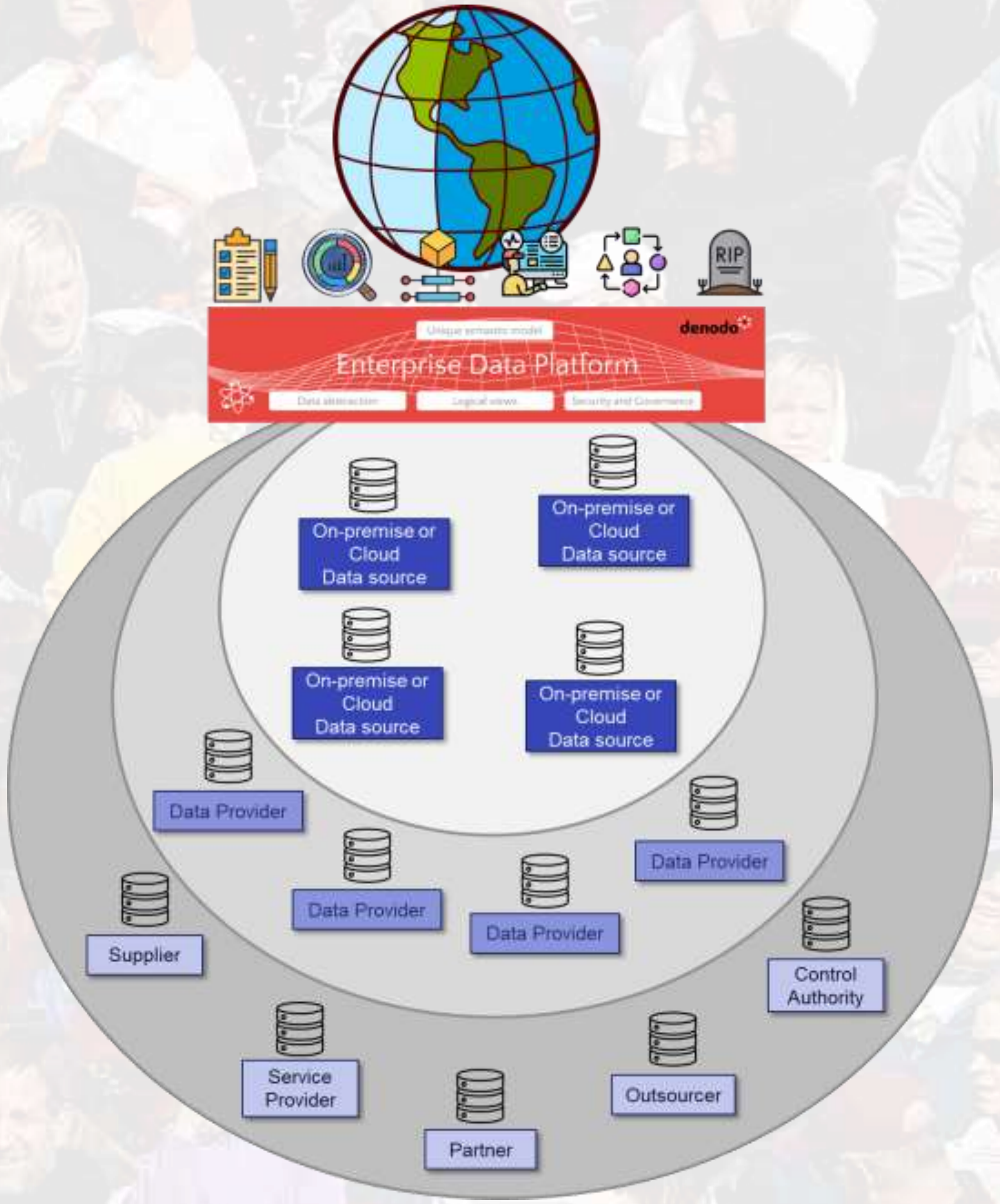
How to use the Augmented Concept View - Examples

Value and management costs of a **physical asset**, such as a building or an industrial plant, which depend on the territorial context in which they are located

Management of **Lifetime Customer Value** and creation of behavioral models capable of predicting its evolution, whether positive or negative

Modelling the **Citizen** throughout his life cycle to anticipate his needs, moving from a reactive to a proactive Public Administration

Develop, produce and market a new **product** or **service**, which involves interaction between different parties throughout all phases of its life cycle



Augmented Views allow a concept to be represented by integrating all the data that enable its precise characterization, regardless of who provides that data, what format it is in, how it is shared, and where it resides

This extension allows you to represent a concept, not only for what it is but also in relation to the context in which it is set and the links that it has with other concepts, going beyond company boundaries, for **data management without barriers**

From Augmented View to the Data Economy

The value of data is also in its ability to circulate freely and within the rules, but the more extensive the space, the clearer the meaning must be

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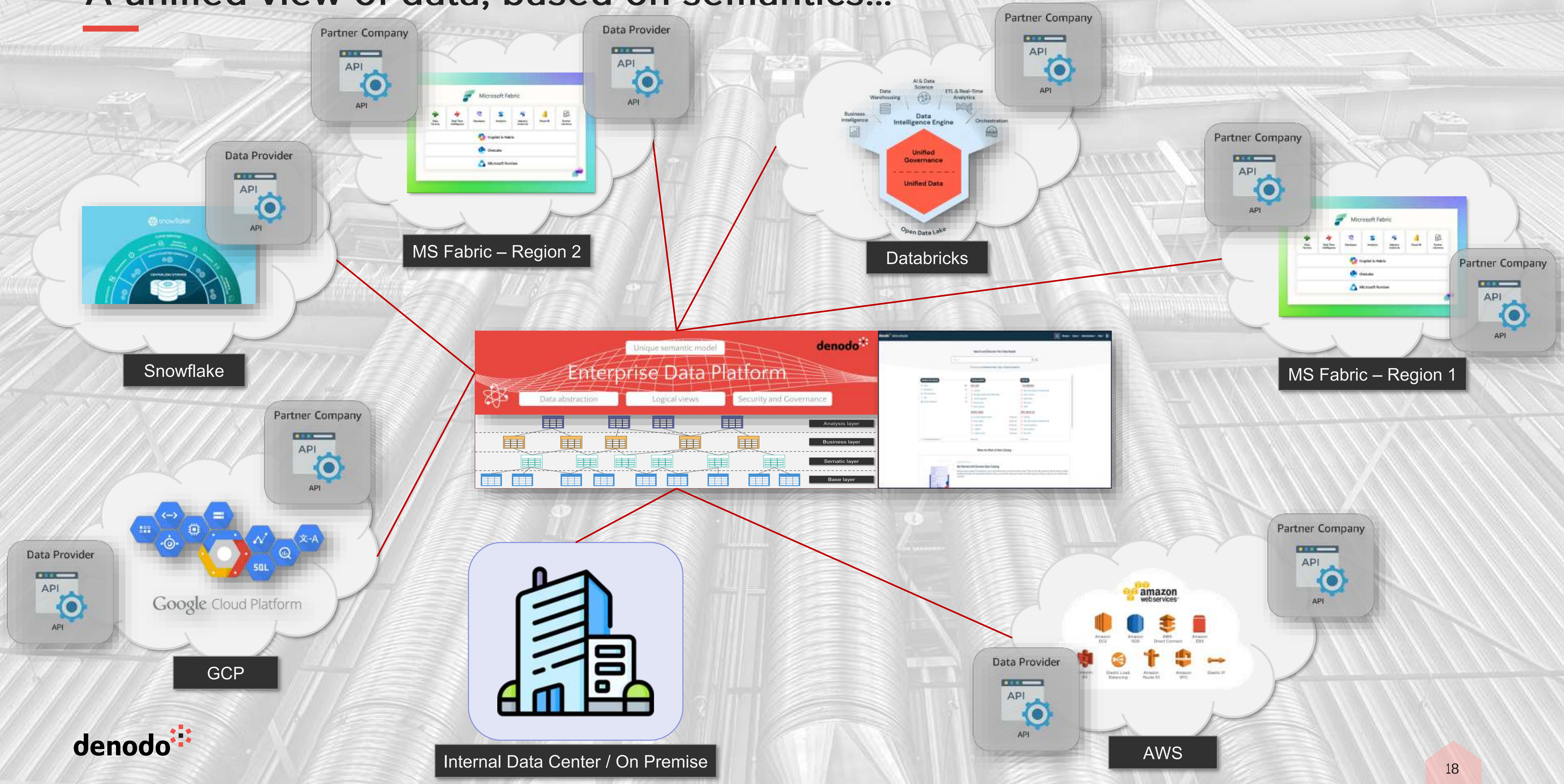
A unified view of data, based on semantics…



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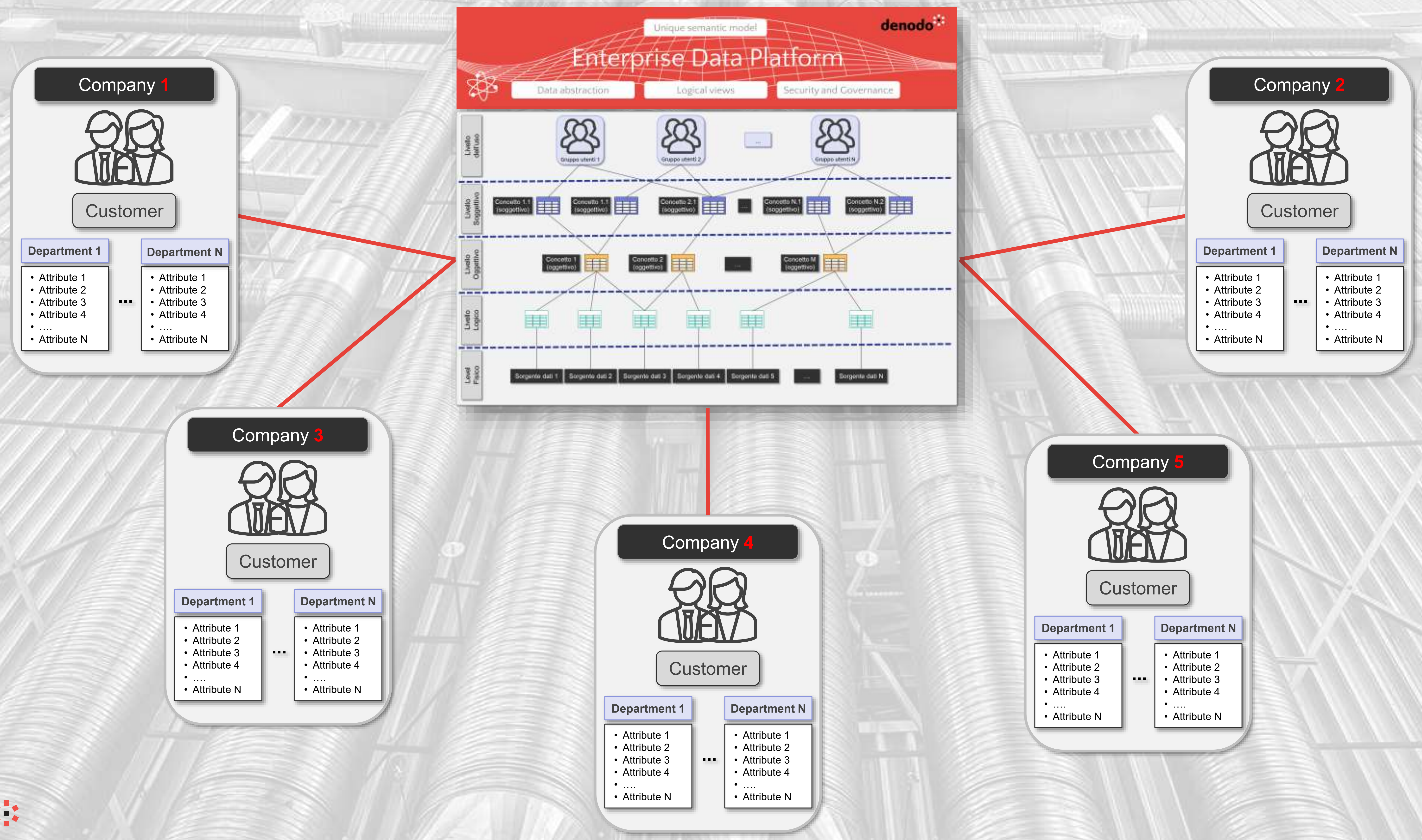
…acting as the glue between subjectivity and objectivity - Example



A formalized semantics, which opens the doors to sharing

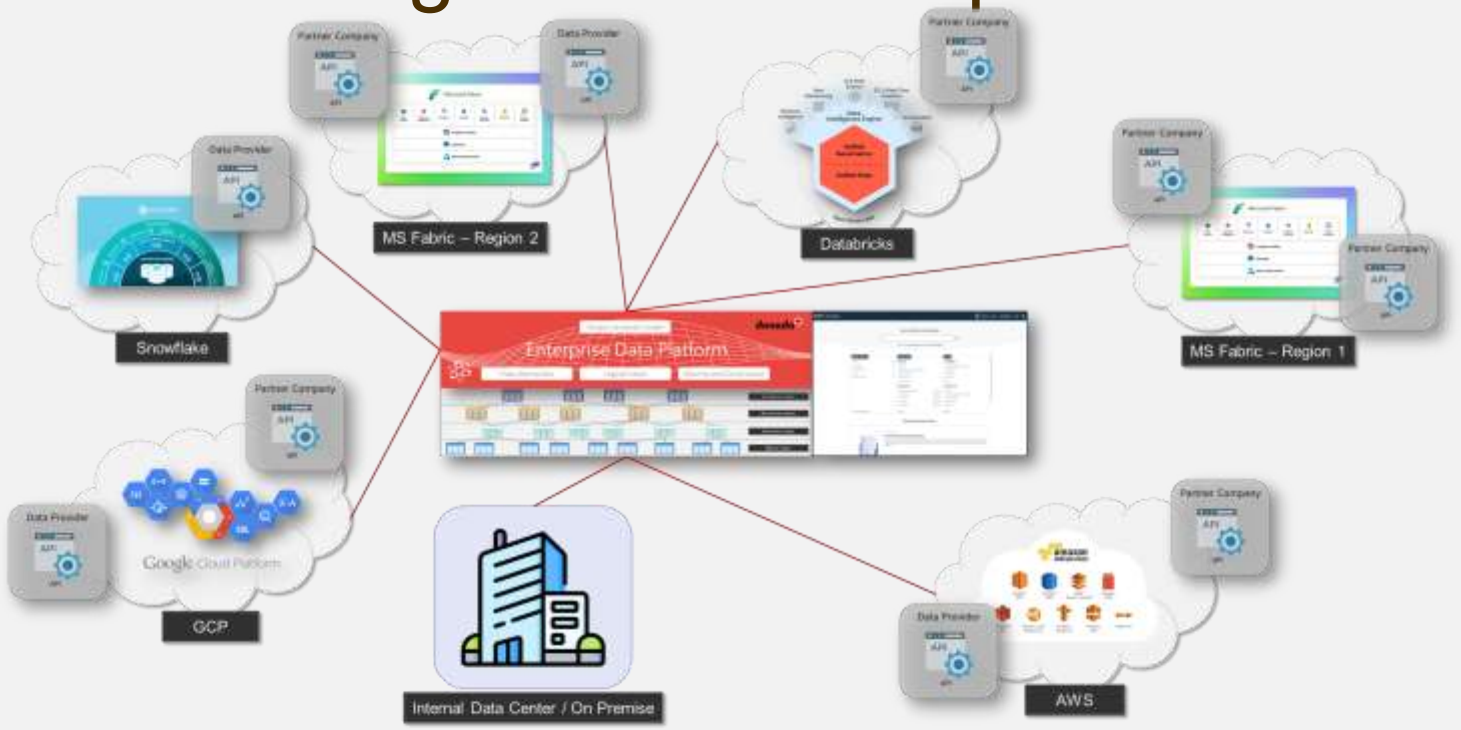
INTERNATIONAL DATA SPACES ASSOCIATION

API + Contract Negotiation

gaia-x

API + Contract Negotiation

Augmented Enterprise



Compliant with

Compliant with

API

Any other Local, National or International Interoperability System

How Virtual Integration Easily Represent a Complex and Detailed Ecosystem

Integrate data, creating a single access point for them and without having to duplicate them beforehand

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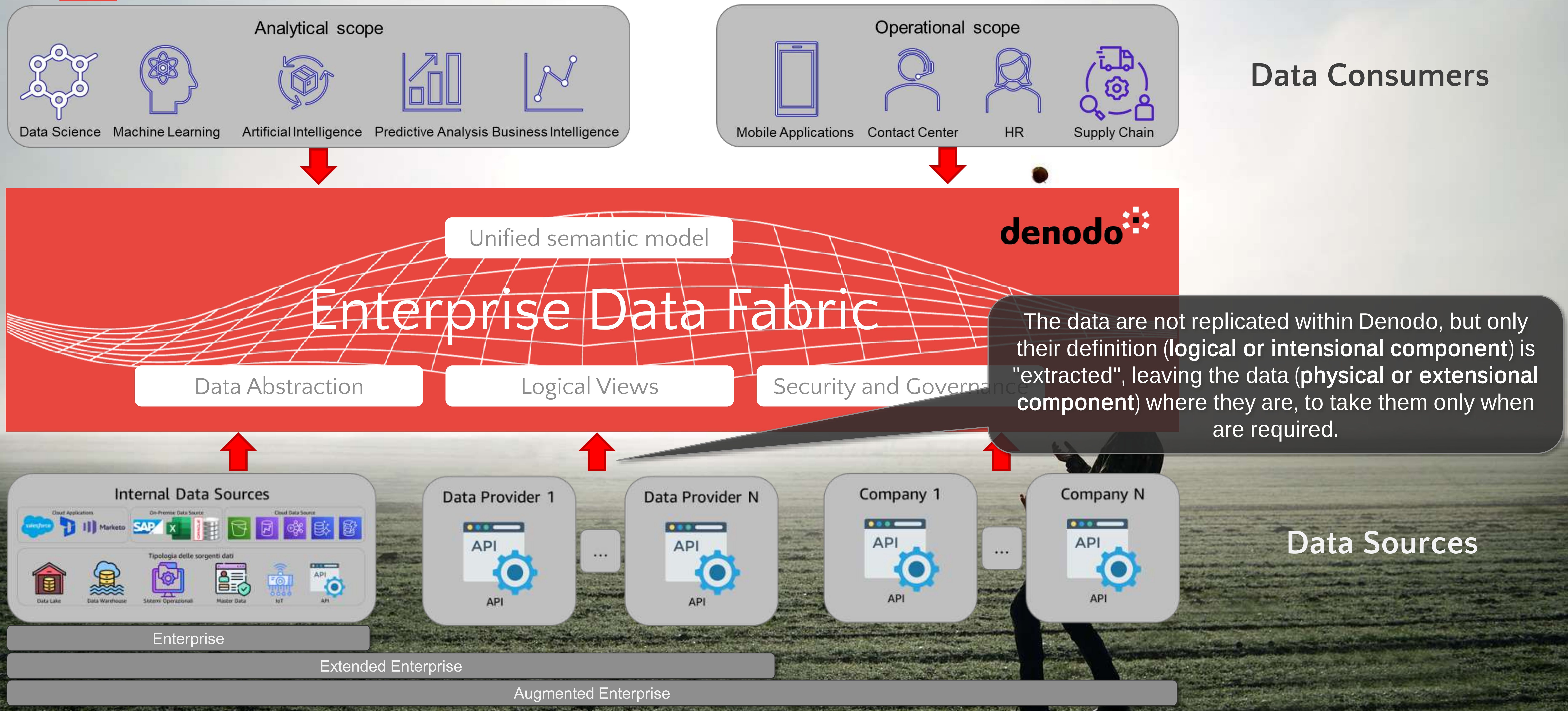
Connect, represent, use



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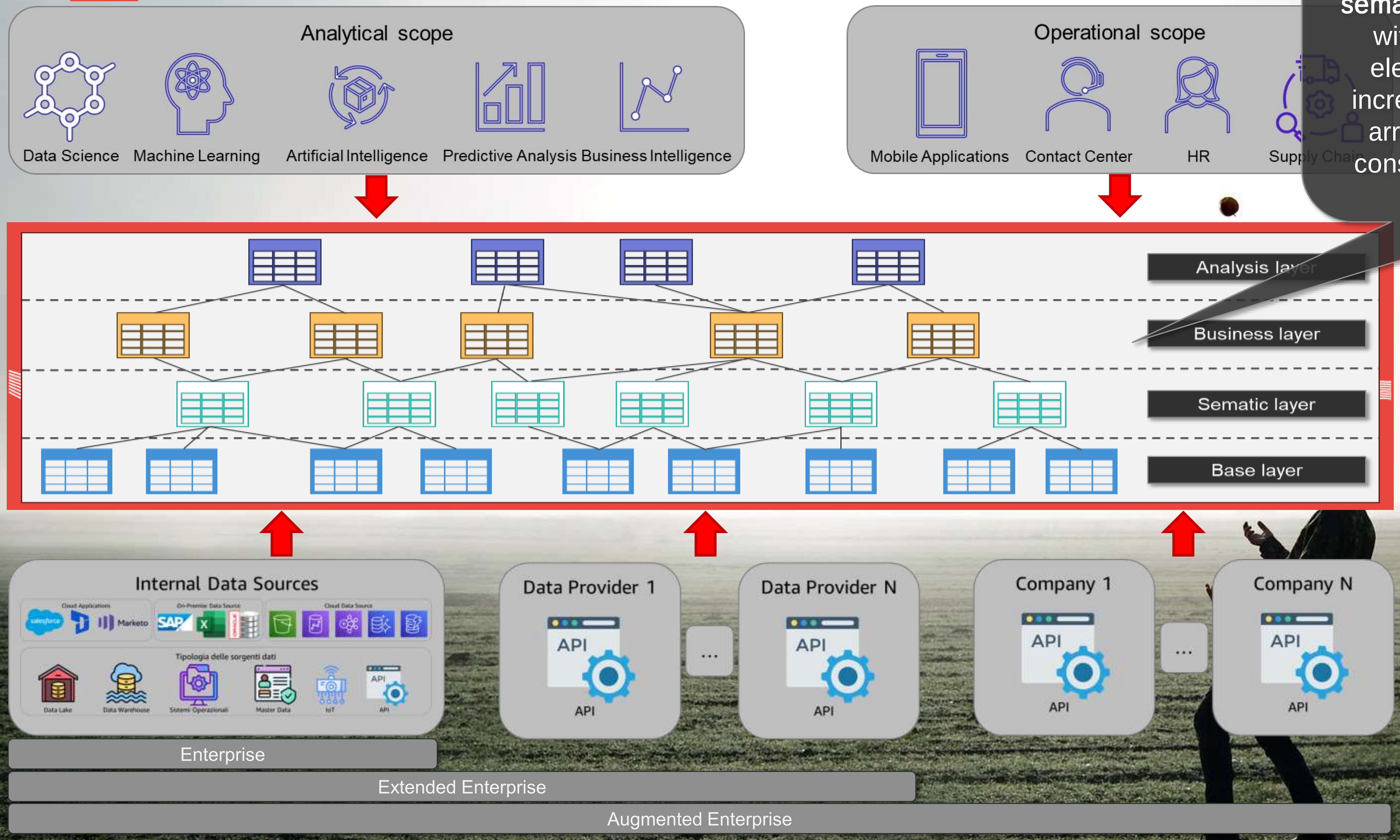
Connect, represent, use

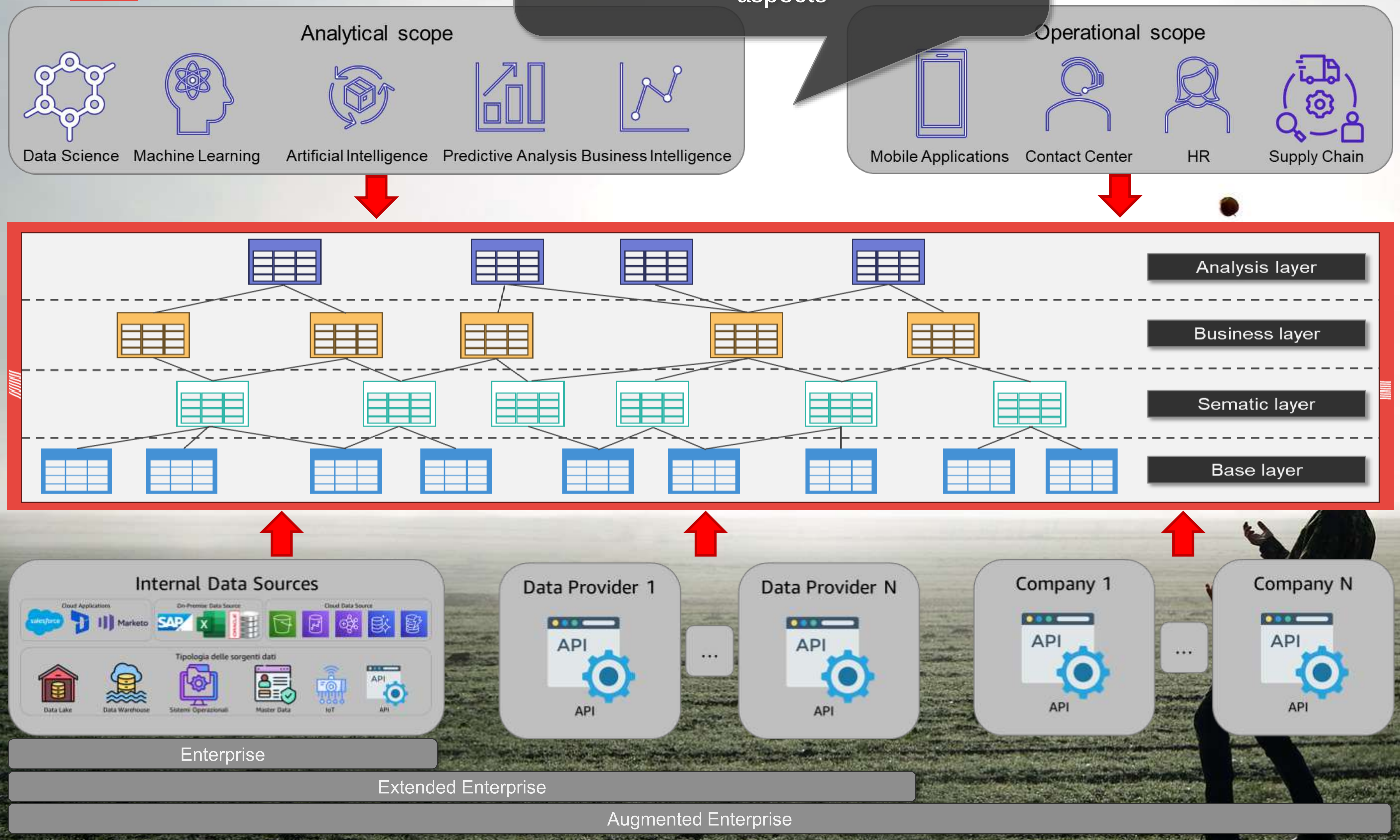
Progressively, as the number of integrated data sources increases, the **semantic model** comes to life, typically with a Lego-Like approach, where elementary data are combined the increasingly articulated ways, so as to arrive at a conceptualization that is consistent with what the organization does



Data Sources

Connect, represent, use

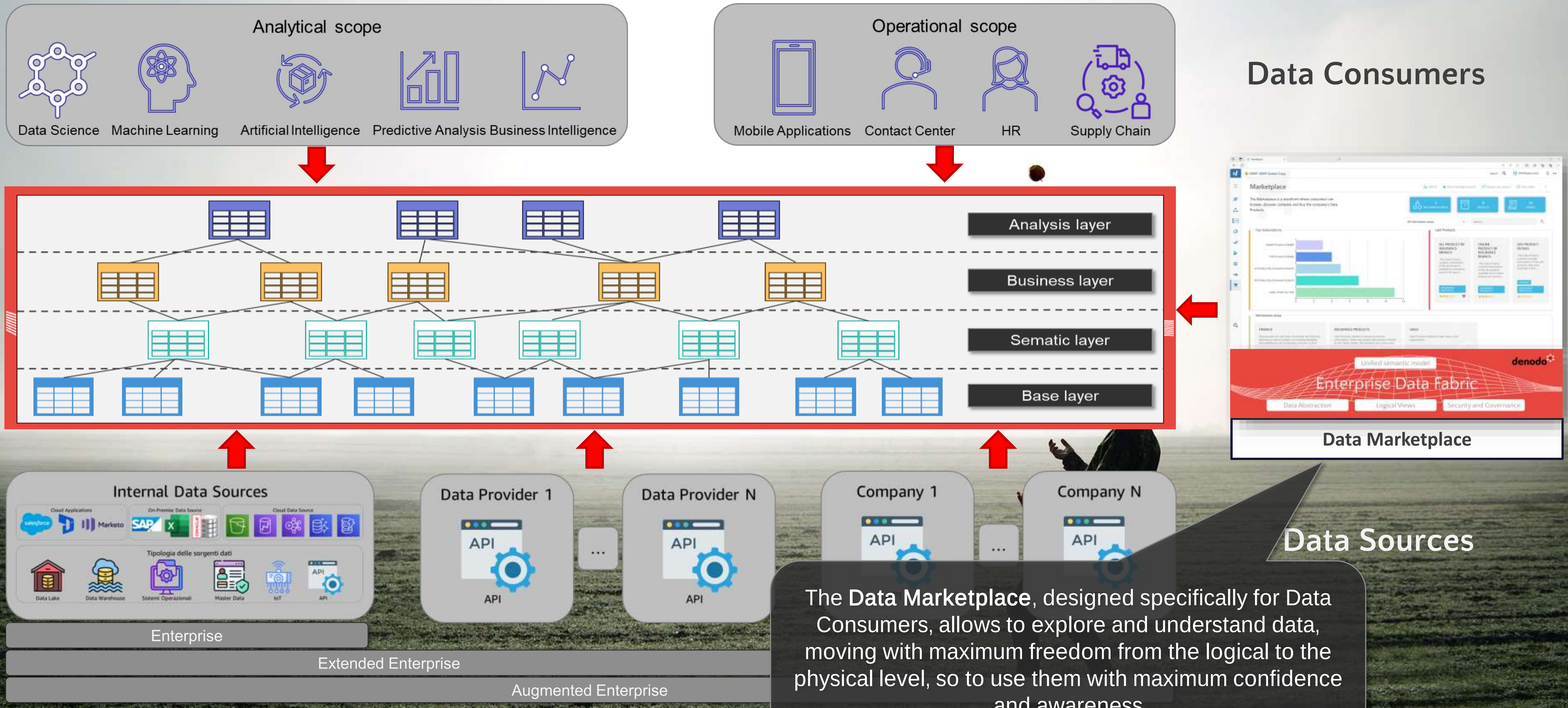
Data consumers can use the data using the tools best suited to their purposes, without having to deal with any technical aspects



Data Consumers

Data Sources

Connect, represent, use



The **Data Marketplace**, designed specifically for Data Consumers, allows to explore and understand data, moving with maximum freedom from the logical to the physical level, so to use them with maximum confidence and awareness

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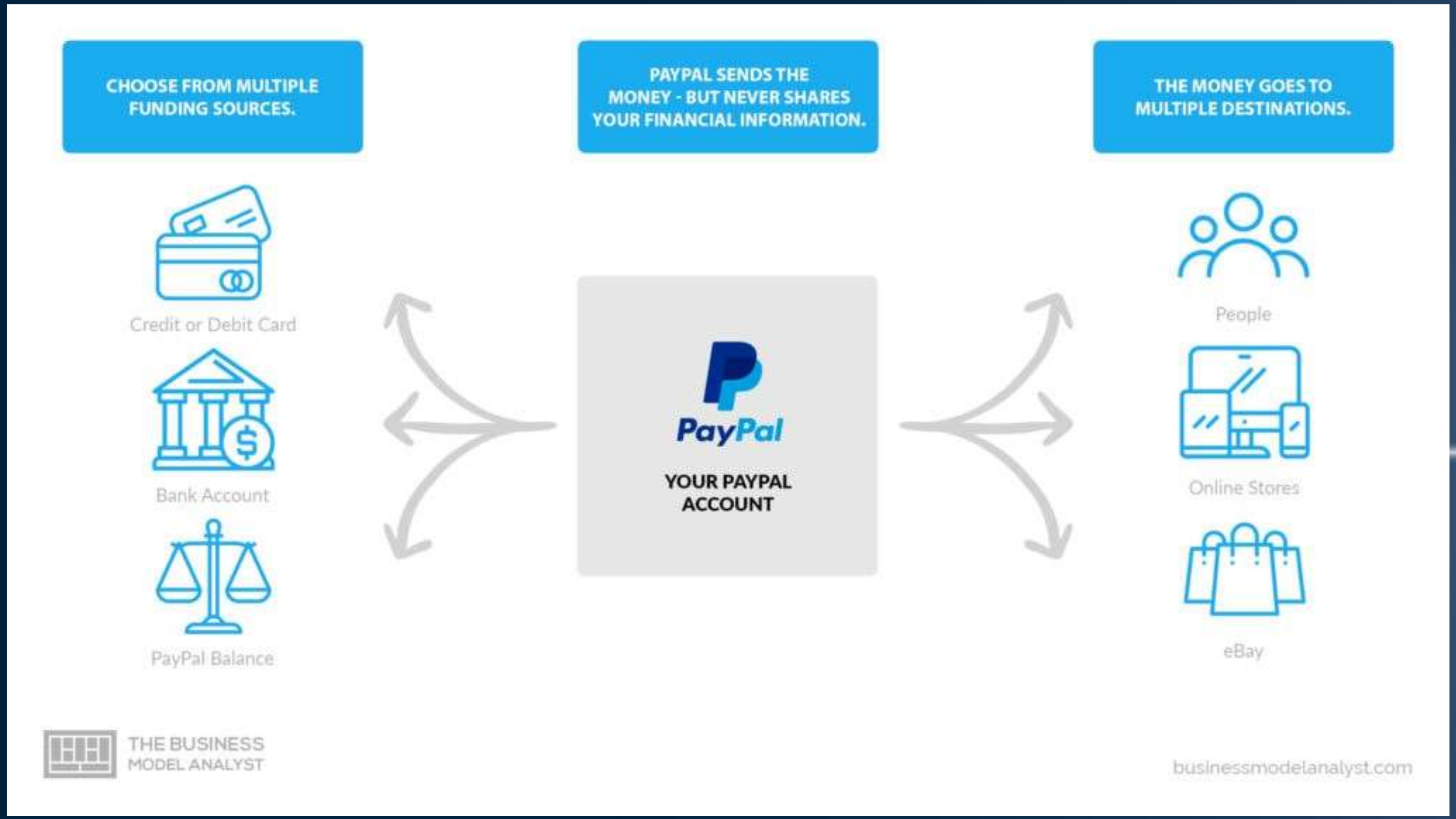
Logical architectures have been among us for a long time...



Our money is not preemptively transferred into PayPal, but can be done if needed

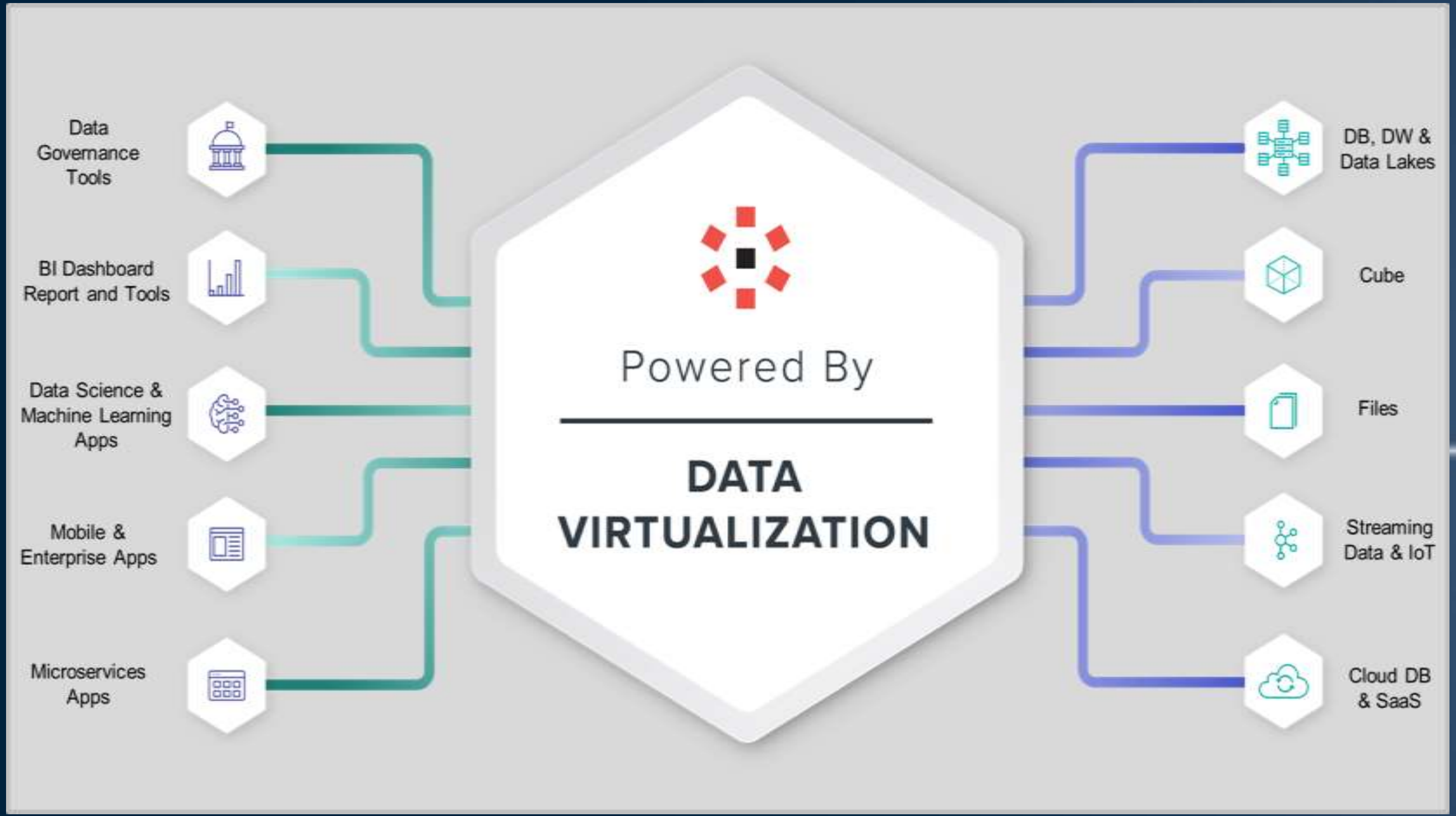
PayPal Maintains a semantic model of the subjects that can send and receive money

Connecting a new payment method is a quick and easy task

If we change credit card or bank account, the change is totally handled by PayPal and Merchants don't notice anything

Security is handled by PayPal and leverages both internal mechanisms and those of our payment methods

Logical architectures have been among us for a long time…



Data are not previously replicated in Denodo, but can be done if needed

Denodo maintains a semantic model of the data that can be accessed and used

Connecting a new data source is a quick and easy task

If we change the technology of a data source, the change is totally managed by Denodo and Data Consumers do not notice anything

Security is managed by Denodo and leverages both internal and data source mechanisms

To conclude

Remember the past to live in the present and be ready for the future

What to take home

Embrace any distributed or not data landscape

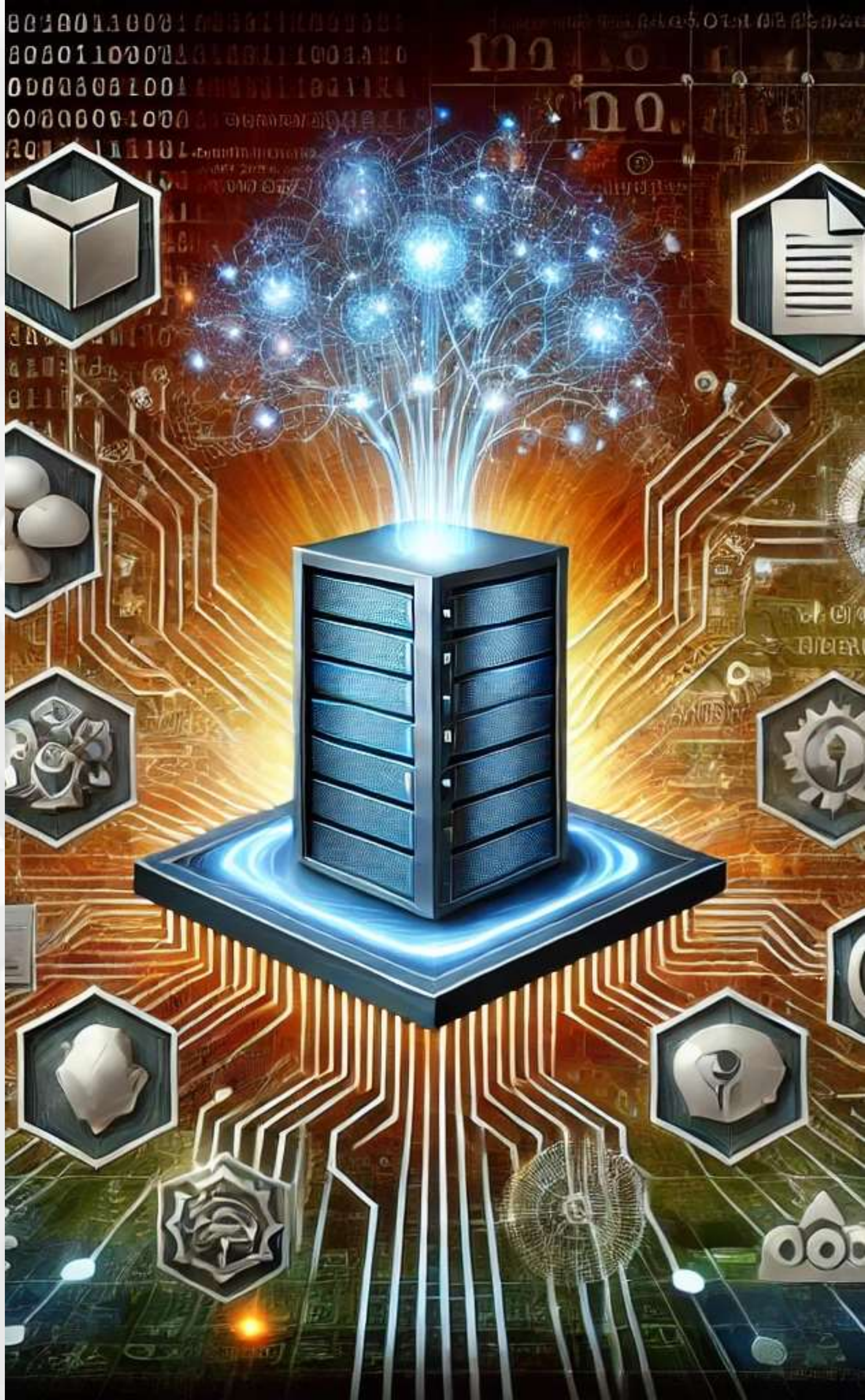
Embrace the fact that data resides in multiple locations or systems – on-prem, hybrid, multi-cloud. All data needs to be managed with consistency

Use a logical approach to manage it

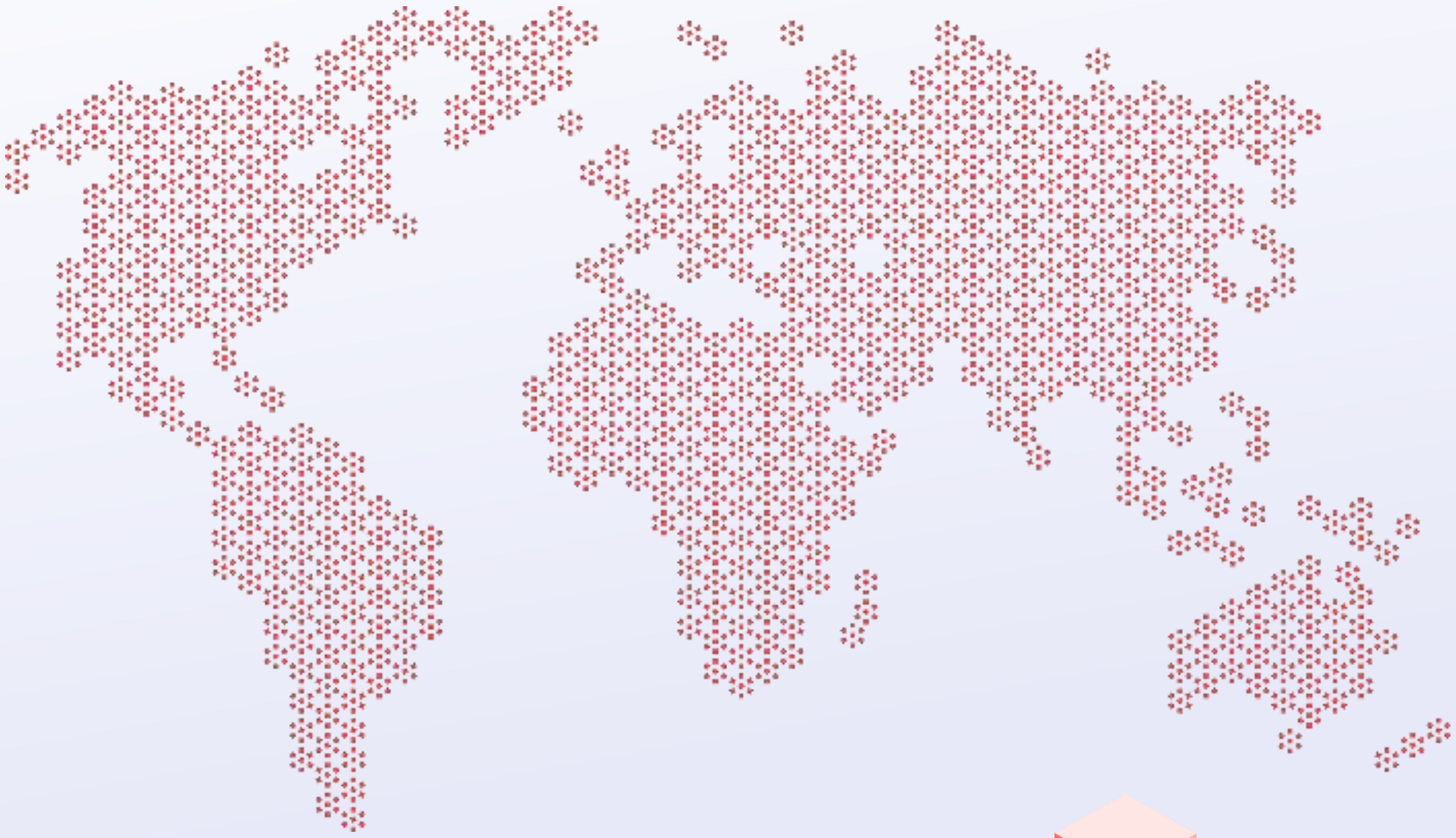
Consumers access data through a centralized semantic model, decoupled from data location and physical schemas, that can enforce security and governance requirements

Adopt AI to simplify and automate

Learn from usage and leverage Large Language Models to automate tedious repetitive tasks



Thanks!



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